

# AI Explainability for Adaptive mmWave Beam Configuration in Dynamic Vehicular Environments

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**Abstract**—Vehicular millimetre-wave (mmWave) networks suffer from beam misalignment and throughput degradation due to high mobility, blockages, and interference—challenges that static beam strategies cannot address. This paper addresses adaptive beam configuration in dynamic vehicular scenarios through reinforcement learning (RL), i.e. Deep Q-Network (DQN) and Q-learning, to jointly optimise base station placement, beam direction, and beamwidth for maximum system sum-rate. Recognising that the lack of interpretability in learned policies remains a critical barrier to operator trust and practical deployment, we integrate a novel multi-agent large language model (LLM) explainability layer that processes user queries in parallel through specialised agents, hence combine numerical logs, scenario rules, and visual outputs into concise, human-understandable explanations. This unified framework demonstrates how high-performance RL optimisation and transparent explainability can be combined to build trust and facilitate practical deployment in future 5G/6G vehicular networks and Open RAN environments.

**Index Terms**—Beam Management, Reinforcement Learning, AI Explainability, LLMs, V2X, O-RAN, 5G/6G.

## I. INTRODUCTION

The evolution toward sixth-generation (6G) wireless networks and Open Radio Access Network (O-RAN) architectures is redefining how intelligence is embedded into communication infrastructures. As artificial intelligence (AI) becomes deeply integrated within the radio access network (RAN)—ushering in the concept of AI-RAN—decision-making is shifting from static, rule-based configurations to adaptive, learning-driven strategies [1], [2]. Within this transformation, millimeter-wave vehicle-to-everything (mmWave V2X) communication has emerged as both a key enabler of ultra-high-speed connectivity and a significant technical challenge, as narrow beams are highly susceptible to misalignment from vehicular mobility, blockages, and dynamic interference.

Traditional beam management techniques—based on exhaustive beam sweeping or static configurations—impose substantial signaling overhead and lack the agility required for

high-mobility environments. To overcome these limitations, recent research has increasingly adopted machine learning (ML) and reinforcement learning (RL) approaches for intelligent, context-aware beam selection and adaptation. Early studies focused on modular subproblems using bandit-based frameworks, [3] introduced a mobility-aware beam selection scheme via multi-armed bandit learning to reduce handover frequency, and later extended this to multi-agent contextual learning for interference management [4]. Dynamic beamwidth adaptation has also shown promise, with contextual bandits enhancing link stability [5] and relay-assisted schemes mitigating blockage effects [6]. However, these efforts primarily target per-link robustness and reliability, rather than optimizing system-wide throughput or coordination.

Building on these foundations, recent advances have explored more sophisticated neural architectures for beam prediction [7], [8]. The authors in [7] employed Liquid Time Constant Neural Networks (LTCNNs), achieving over 94% reduction in beam training overhead for vehicle-to-infrastructure (V2I) scenarios. [8] developed a multi-agent deep RL framework named DeepBeam, incorporating transformer-based trajectory prediction that sustains very high beam alignment accuracy at vehicular speeds up to 120 km/h. Recognising the scarcity of real-world training data, [9] mapped physical environments into synchronised digital counterparts to enable efficient beam tracking through digital twin (DT) technologies. [10] proposed a sim-to-real meta-learning framework using multimodal synthetic datasets, achieving accuracy gains exceeding an order of magnitude.

Despite these advances in beam prediction and throughput optimisation, one critical dimension remains largely unaddressed: explainability. In safety-critical vehicular contexts and regulated O-RAN deployments, operators must understand why an AI-driven system makes particular decisions to ensure trust, accountability, and safety compliance. The broader field of

Explainable Artificial Intelligence (XAI) has established multi-dimensional frameworks for achieving this transparency [11]. In addition, [12] highlighted the potential of large language models (LLMs) as natural interfaces for explainability, translating complex algorithmic behaviors into coherent human-understandable narratives. However, the integration of LLM-driven explainability into mmWave beam management systems remains unexplored.

To address this gap, we propose a system-level framework that jointly optimizes base station placement, beam direction, and beamwidth for throughput maximization while providing interpretable, evidence-based reasoning. Our approach integrates reinforcement learning with an LLM-driven explainability layer composed of specialized SQL, Vector, and Visual agents that feed structured evidence into an Analysis-AI module synthesizing transparent explanations. We evaluate two RL strategies—Deep Q-Networks (DQN) for rapid convergence and factorized tabular Q-learning for superior asymptotic performance—demonstrating that coupling RL optimization with LLM-based explainability achieves both high performance and interpretability, establishing a pathway toward trustworthy vehicular O-RAN systems.

The remainder of this paper is organised as follows. Section II presents the system model, including the simulation environment and formulation of the beam configuration problem. Section III details the proposed AI-driven explainability layer, describing the multi-agent architecture. Section IV presents experimental results on the explainability layer. Finally, Section V concludes with key findings and outlines future research directions for integrating explainable AI into O-RAN-based vehicular communication systems.

## II. SYSTEM MODEL

### A. Scenario Setup

To create a realistic vehicular communication scenario for testing beam steering strategies, we design a custom urban mobility and wireless propagation simulator. The environment represents a typical city intersection with dynamic traffic patterns and realistic radio propagation characteristics. We simulate a  $500\text{ m} \times 500\text{ m}$  urban environment containing a sparse two-way road grid and a central roundabout, as illustrated in Fig. 1. The environment comprises horizontal and vertical road segments with varying speed limits (30 or 50 km/h) and a roundabout centered at (125 m, 125 m) with a 25 m radius, 6 m lane width, and 20 km/h speed limit. Three axis-aligned rectangular blockages are placed in the environment to create realistic LOS/NLOS transitions. Each blockage is defined by its spatial extent ( $x$ -range and  $y$ -range in meters) and a multiplicative attenuation factor that reduces signal strength when rays penetrate through it. Specifically, blockage B1 spans from 40 m to 80 m in the  $x$ -direction and 135 m to 160 m in the  $y$ -direction with an attenuation factor

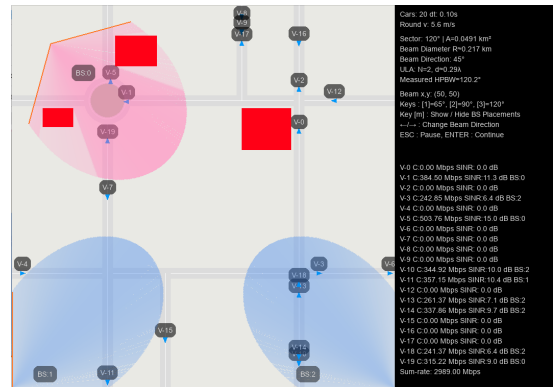


Fig. 1: Visualization of the simulation topology and execution showing the 500 m  $\times$  500 m urban environment with road segments, blockages, base station coverage patterns, vehicle trajectories and beam steering dynamics over time.

of 0.5, blockage B2 occupies 300–365 m (x) and 135–190 m (y) with attenuation 0.2, and blockage B3 covers 135–190 m (x) and 40–80 m (y) with attenuation 0.5. A population of  $N = 20$  autonomous vehicles circulates along the road network at fixed per-segment speed limits. Vehicles spawn at map-edge entry points and are reinserted at randomly selected edges to maintain constant load, following lane-center trajectories without acceleration, collisions, or car-to-car interactions. Time is discretized with  $\Delta t = 0.1$  s, and each episode lasts up to 1,800 steps (180 s).

The wireless configuration includes one controllable mmWave base station whose beam parameters will be learned through reinforcement learning, along with two sources of interference that remain fixed throughout training. The wireless scene comprises one controllable serving sector (the RL agent) and two fixed interfering base stations placed at the lower corners of the map, each with  $90^\circ$  fixed orientations pointed inward. The optimization objective is to evaluate 75 candidate base station placements distributed across three segments (Left/Center/Right, with 25 m spacing between neighbors) and, for each location, learn the transmit beam direction and beamwidth that maximize network utility defined as the sum of per-UE capacities. The radio propagation model follows 3GPP specifications for mmWave frequencies, incorporating path loss, antenna patterns, and realistic link budget parameters. The system operates on the n258 band (24.25–27.5 GHz) with carrier frequency  $f_c = 26$  GHz and bandwidth  $BW = 100$  MHz. Propagation follows the Close-In (CI) path loss model with LOS/NLOS exponents. The path loss in decibels is given by:

$$\text{PL}(d)[\text{dB}] = \text{FSPL}(f_c, d_0) + 10n \log_{10} \left( \frac{d}{d_0} \right) + X_\sigma, \quad (1)$$

where  $\text{FSPL}(f_c, d_0)$  is the free-space path loss at reference distance  $d_0 = 1$  m,  $n$  is the path loss exponent (different for LOS and NLOS conditions),  $d$  is the transmitter-receiver separation distance, and  $X_\sigma$  represents shadow fading. The received signal power at a UE is computed from the link budget equation:

$$P_{\text{rx}}[\text{dBm}] = \text{EIRP}[\text{dBm}] + G_{\text{rx}}[\text{dBi}] - \text{PL}(d)[\text{dB}] - L_{\text{misc}}[\text{dB}] - L_{\text{blockage}}[\text{dB}], \quad (2)$$

where  $\text{EIRP} = 55$  dBm is the effective isotropic radiated power (including array-factor gain from beamforming),  $G_{\text{rx}} = 8$  dBi is the UE receive antenna gain,  $L_{\text{misc}} = 15$  dB accounts for miscellaneous losses (body loss, penetration, etc.), and  $L_{\text{blockage}}$  is the additional attenuation when the signal passes through blockages. The noise power is calculated as:

$$P_{\text{noise}}[\text{dBm}] = -174 + 10 \log_{10}(\text{BW}[\text{Hz}]) + \text{NF}[\text{dB}], \quad (3)$$

where  $\text{NF} = 7$  dB is the receiver noise figure. The transmit pattern is derived from a ULA array factor with Hamming taper, ensuring the  $-3$  dB width matches the selected HPBW. A sidelobe floor of  $-13$  dB prevents unrealistic nulls, and under these settings, a fixed beam covers approximately  $0.0491 \text{ km}^2$ . Each UE connects to the strongest beam, while interference is aggregated from the top- $K = 3$  non-serving beams. Per-UE SINR is calculated against the three strongest interferers, and capacity follows Shannon's formula:

$$C = \text{BW} \cdot \log_2(1 + \text{SINR}). \quad (4)$$

### B. Problem Formulation and Learning

Having established the simulation environment and radio propagation characteristics, we now formulate the beam steering problem as a Markov Decision Process (MDP) suitable for reinforcement learning. The MDP consists of a discrete action space representing beam configurations, a compact observation space capturing traffic patterns, and a reward function that balances network performance with operational constraints. We formally define MDP as a tuple  $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ , where:

**State Space  $\mathcal{S}$ :** The state  $s_t \in \mathcal{S}$  at time step  $t$  consists of two components: (1) a discretised vehicle-density grid  $\mathbf{G}_t \in N^{m \times m}$  where each cell contains the count of vehicles within a  $20 \text{ m} \times 20 \text{ m}$  spatial bin, and (2) the current beam configuration  $b_t = (\theta_t, \omega_t)$  where  $\theta_t$  is the azimuth index and  $\omega_t$  is the beamwidth index. Thus,  $s_t = (\mathbf{G}_t, \theta_t, \omega_t)$ .

**Action Space  $\mathcal{A}$ :** Action  $a_t \in \mathcal{A}$  represents the beam configuration to apply at time step  $t$ . The agent selects from 24 discrete beam configurations formed by 8 azimuths in  $45^\circ$  steps  $\Theta = \{0^\circ, 45^\circ, \dots, 315^\circ\}$  crossed with 3 half-power beamwidths  $\Omega = \{65^\circ, 90^\circ, 120^\circ\}$ . Therefore,  $\mathcal{A} = \Theta \times \Omega$  with  $|\mathcal{A}| = 24$ .

**Reward Function  $\mathcal{R}$ :** The reward  $r_t = \mathcal{R}(s_t, a_t, s_{t+1})$  quantifies the marginal gain in total network capacity attributable to the serving beam, with penalties to discourage excessive beam reconfiguration. Let  $C_t^{\text{srv}}$  denote the sum capacity with the serving beam active, and  $C_t^{\text{int}}$  the sum capacity with only interfering BSs:

$$C_t^{\text{int}} = \sum_{u \in \mathcal{U}_t} C_u \text{ (interfering BSs only)}, \quad (5)$$

$$C_t^{\text{srv}} = \sum_{u \in \mathcal{U}_t} C_u \text{ (serving BS active)}. \quad (6)$$

The step reward is then:

$$r_t = \frac{C_t^{\text{srv}} - C_t^{\text{int}}}{\Delta t} - \lambda_\theta \Delta_\theta(a_t, a_{t-1}) - \lambda_\omega \Delta_\omega(a_t, a_{t-1}), \quad (7)$$

where  $\Delta_\theta$  counts the number of  $45^\circ$  azimuth hops,  $\Delta_\omega$  indicates beamwidth changes, and the penalty coefficients are  $\lambda_\theta = 200$  Mbps and  $\lambda_\omega = 100$  Mbps. The policy  $\pi(a|o)$  aims to maximize the expected episodic return:

$$J(\pi) = E \left[ \sum_{t=0}^{T-1} \gamma^t r_t \mid \pi \right], \quad (8)$$

with discount factor  $\gamma = 0.99$  and horizon  $T$  up to the episode cap (1,800 steps).

We employ two RL models with different approaches that represent distinct points in the sample-efficiency versus asymptotic-performance tradeoff for solving the formulated MDP. These two reinforcement learning approaches are compared under identical episode budgets and evaluation conditions. The first is a Deep Q-Network (DQN) that approximates the action-value function using a neural network. DQN learns the Q-function  $Q(s, a; \theta)$  parameterized by weights  $\theta$ , which is updated by minimizing the temporal difference error:

$$\mathcal{L}(\theta) = E_{(s, a, r, s') \sim \mathcal{D}} \left[ \left( r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta) \right)^2 \right], \quad (9)$$

where  $\mathcal{D}$  is the experience replay buffer and  $\theta^-$  are the target network parameters updated periodically. Our implementation uses Stable-Baselines3 with MlpPolicy, experience replay (buffer size  $10^5$ ), target network updates, batch size 256, and  $\epsilon$ -greedy exploration decaying linearly from 1.0 to 0.05 over training. The second is a tabular Q-learning method with a factorized state representation that reduces the density grid to coarse bins of center-of-mass and occupancy, with a single Q-table shared across all base station placements to exploit structural similarities. Both learners employ early stopping based on a rolling-window average reward criterion. Table I summarizes all key simulation parameters.

TABLE I: Key parameters of the simulation setup.

| Category     | Parameter         | Value / Setting                                                                            |
|--------------|-------------------|--------------------------------------------------------------------------------------------|
| Environment  | Map size          | 500 m $\times$ 500 m                                                                       |
|              | BS placements     | 75 candidate locations (25 m spacing)                                                      |
|              | Blockages         | B1 (40–80, 135–160, 0.5), B2 (300–365, 135–190, 0.2), B3 (135–190, 40–80, 0.5)             |
| Traffic      | Time step         | $\Delta t = 0.1$ s                                                                         |
|              | Episode length    | $\leq 1,800$ steps (180 s)                                                                 |
|              | Vehicles          | $N = 20$ (edge spawn, kinematic motion)                                                    |
|              | Roundabout        | $r = 25$ m, single lane, 20 km/h                                                           |
|              | Road speeds       | 30/50 km/h (segment-dependent)                                                             |
| Beam Config  | Agent beam        | 1 steerable sector (azimuth + HPBW)                                                        |
|              | Interfering BSs   | 2 fixed sectors (map corners, $90^\circ$ )                                                 |
|              | Azimuth choices   | $\{0^\circ, 45^\circ, \dots, 315^\circ\}$ (8 directions)                                   |
|              | Beamwidth choices | $\{65^\circ, 90^\circ, 120^\circ\}$ (3 widths)                                             |
| Observation  | Density grid      | 20 m $\times$ 20 m bins, vehicle counts                                                    |
|              | Beam state        | (azimuth index, beamwidth index)                                                           |
| Action Space | RL actions        | Discrete-24 ( $8 \times 3$ )                                                               |
| Reward       | Marginal gain     | $\Delta$ (sum-rate with agent vs. interference-only)                                       |
|              | Movement penalty  | 200 Mbps per $45^\circ$ azimuth; 100 Mbps per width change                                 |
| RF Model     | Carrier           | n258, $f_c = 26$ GHz, BW = 100 MHz                                                         |
|              | Path loss         | Close-In model (LOS/NLOS, 1 m reference)                                                   |
|              | Link budget       | EIRP = 55 dBm, $G_{rx} = 8$ dBi, losses = 15 dB, NF = 7 dB                                 |
|              | Capacity          | Shannon: $C = BW \cdot \log_2(1 + \text{SINR})$                                            |
| Interference | Top- $K$ model    | Sum of 3 strongest non-serving beams                                                       |
|              | DQN (SB3)         | MlpPolicy, $\gamma = 0.99$ , buffer = $10^5$ , batch = 256, $\epsilon: 1 \rightarrow 0.05$ |
| Learning     | Tabular Q         | Factorized state, shared Q-table across placements                                         |
|              | Early stopping    | Rolling-window mean reward threshold                                                       |

For enhanced interpretability of the learned policies, we complement the RL controllers with an LLM-based explainability layer that translates numerical outcomes into human-understandable narratives. The RL controllers are complemented by an LLM-based explainability layer that interprets learning outcomes and beam decisions. User queries are processed in parallel by three specialized agents: a Visual-AI agent analyzing figures such as learning curves and heatmaps, an SQL-AI agent retrieving quantitative metrics from the run database, and a Vector-AI agent fetching contextual information including simulation rules, RF models, and reward design. An Analysis-AI agent then integrates these numerical, contextual, and visual inputs into human-understandable narratives. This multi-agent pipeline ensures that explanations remain grounded in recorded metrics and documented assumptions, building transparency and trust for deployment in real vehicular O-RAN scenarios.

### III. PROPOSED AI-DRIVEN EXPLAINABILITY LAYER

#### A. Overview

We designed a multi-agent LLM reasoning layer on top of the mmWave RL simulator (DQN vs. factorised Q-learning) using n8n as the orchestrator (Fig. 2 actionable explanations of RL results). A SQL-AI agent queries a structured log data (rewards, actions), a Vector-AI agent retrieves simulation rules (reward shaping, RF model, constraints), and an Visual-AI agent interprets plots/screenshots. Analysis-AI agent composes a three-part answer—Summary, Interpretation, Recommendations—prioritising numeric evidence (SQL), domain reasoning

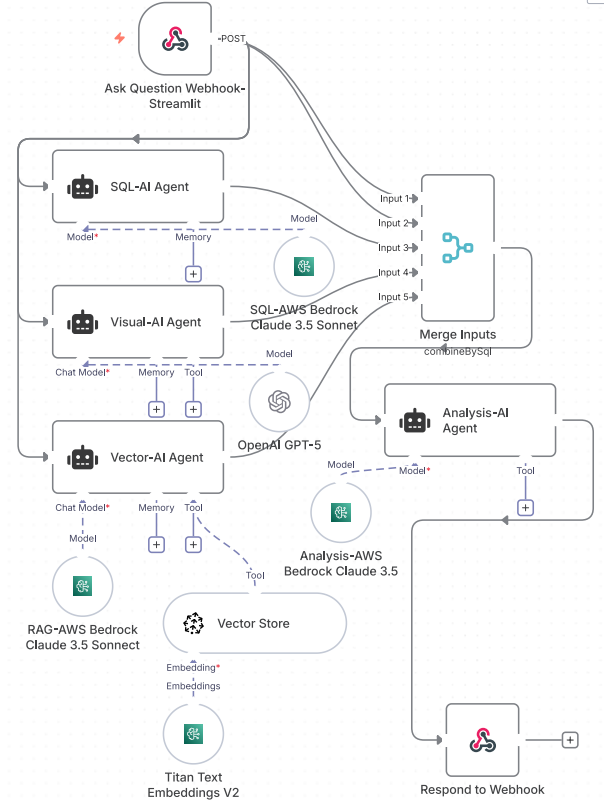


Fig. 2: Workflow of the LLM-based Explainability Pipeline.

(RAG), and visual trends (images). This design ensures that explanations are grounded in recorded metrics and documented assumptions, not free-form speculation, and that explanations of simulation results are reported under consistent budgets and seeds.

The system fuses three evidence sources:

- **Numerical Evidence (SQL)** — concrete metrics from a Postgres database of runs, rewards and top actions.
- **Domain Rules/Context (Vector/RAG)** — simulation assumptions, reward design, RF model, geometry, etc., embedded and indexed in a ElasticSearch.
- **Visual Evidence (Visual Agent)** — analyses of uploaded snapshots (learning curves, heatmaps, HUD screenshots).

Analysis-AI Agent composes a structured answer with three sections—*Summary*, *Interpretation*, and *Recommendations*—while enforcing that claims are grounded in concrete metrics or documented assumptions.

#### B. LLM End-to-End Flow

The explainability workflow begins when the user submits a query, together with optional images, through the UI as illustrated in Fig. 3 This input is routed via a webhook and

## Explanation of Simulation Results

Regarding to algorithm graphs in the images, why DQN is faster than Q-Learning in convergence?

### Summary

- DQN achieved a higher average final evaluation reward (119.34) compared to Q-Learning (112.21) over 75 runs each.
- Training curves (from image analysis) show that one algorithm (presumed DQN) reaches a high-reward plateau early (~100-200 episodes) with higher variance, while the other (presumed Q-Learning) improves more slowly and steadily across all 769 episodes, reaching lower final rewards.
- This indicates DQN converges faster and reaches better final performance, although with more variability.

### Interpretation

DQN's faster convergence is due to several function approximation advantages:

- DQN uses neural networks to generalize from one experience to many similar states, unlike Q-Learning's one-at-a-time tabular updates. This speeds up learning across large or continuous spaces.
- Experience replay allows DQN to reuse past transitions more effectively per update, improving sample efficiency.
- The target network in DQN stabilizes learning by reducing feedback oscillations, helping it climb to high performance quickly and maintain it.
- In the visual plots, the steep early rise in the DQN-like curve reflects quick policy discovery that avoids movement penalties (e.g. costly 45° hops or sector switches) while leveraging capacity gain. Q-Learning shows steady improvement, suggesting slower adaptation to these reward trade-offs.

### Recommendations

- Plot counts of 45° hops and sector switches per episode for both algorithms to confirm whether DQN discovers low-penalty behavior earlier.
- Test removing DQN components ( replay buffer, target networks) to confirm their role in accelerating convergence.
- Standardize hyperparameters (e.g., learning rates, update schedules) to isolate whether architecture, not configuration, drives the speed difference.

Your question...

Fig. 3: AI Chatbot UI

processed in parallel by three specialised AI agents in n8n LLM pipeline. The *Visual Agent* analyses uploaded figures such as learning curves or heatmaps and converts them into structured observations; the *SQL Agent* generates a deterministic, single Postgres query to extract quantitative metrics such as mean rewards, action frequencies, and positional data; and the *Vector Agent* retrieves domain-specific context from scenario documentation, including RF assumptions, penalty terms, and action-space constraints. Their outputs are then forwarded to the *Analysis-AI Agent*, which merges numerical, contextual, and visual evidence into a structured three-part explanation consisting of a *Summary*, *Interpretation*, and *Recommendations*. Finally, the merged response is returned to the UI as a human-readable output. This architecture ensures a clear flow from user input, through parallel multi-agent reasoning, to evidence-grounded explanations delivered back to the operator.

### C. Outputs and Explainability

The merged response is presented as:

- Summary:** Key numerical metrics (e.g., mean rewards, action frequencies).
- Interpretation:** Explanations derived from domain rules and visual evidence, explaining why the observed results occur.
- Recommendations:** Suggested next steps, such as increasing seeds, performing ablation on penalty terms, or aligning budgets across algorithms.

This approach enforces evidence-gated explainability: the system cannot speculate freely—numbers must originate from SQL logs, contextual reasoning from RAG, and qualitative trends from images. In this way, explanations are both reproducible and counterfactual-friendly. By integrating numerical, contextual, and visual evidence, this explainability layer ensures that AI interpretations remain faithful to the simulation, offering transparency for comparing algorithms such as DQN and factorised Q-learning under consistent experimental con-

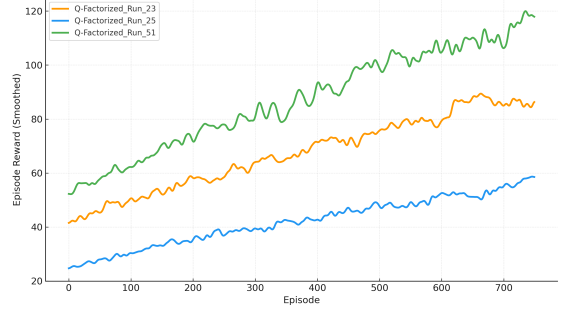


Fig. 4: Q-Factorised episode reward comparison across locations

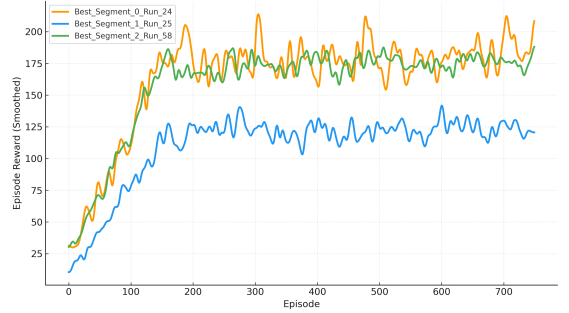


Fig. 5: DQN Episode reward comparison across locations

ditions.

## IV. RESULTS AND DISCUSSION

In this section we present the comparative performance of the two reinforcement learning approaches—factorised Q-learning and DQN—and discuss the role of the LLM-based explainability layer in interpreting the outcomes. Fig. 4 and 5 plot the learning curves of both methods under equal interaction budgets at different locations.

### A. RL Performance Comparison

The results show a clear trade-off between the two families of controllers. The DQN agent exhibits rapid improvement in marginal sum-rate during the early stages of training, reflecting its strength in sample efficiency and fast convergence under limited interaction budgets. In contrast, the factorised Q-learning agent progresses more slowly but achieves higher asymptotic performance when trained over larger budgets. This indicates that exploiting coarse structural features of the vehicular density grid allows the tabular method to discover policies that generalise better in the long run, at the cost of slower early adaptation.

Another important observation is the impact of penalty terms on policy stability. Both methods learn to balance throughput gains with beam switching costs, but the tabular approach tends to yield more stable beam selections once converged,

whereas DQN occasionally explores aggressive steering that introduces transient fluctuations. These findings underscore the importance of evaluating RL controllers not only in terms of final sum-rate, but also stability, switching frequency, and operational realism.

### B. Explainability of Results

Beyond raw performance metrics, the integration of the LLM-based explainability layer provides crucial interpretability. For example, the Visual-AI agent highlights that DQN's early advantage comes from quickly locking onto wide beams (e.g.,  $90^\circ$  or  $120^\circ$ ) that cover high-density traffic segments, while the SQL-AI agent confirms a higher frequency of azimuth switching in those runs. The Vector-AI agent contextualises this behaviour with simulation rules, noting that penalties of 200 Mbps per azimuth hop discourage excessive steering in the long term. The Analysis-AI agent combines these inputs into narratives such as: "DQN achieves faster initial growth by exploiting wide beams but incurs penalties from frequent switching, whereas tabular Q-learning converges more slowly yet maintains steadier beam configurations."

### C. Discussion

The results show that reinforcement learning can outperform static beam strategies by adapting to vehicular mobility and interference, though the optimal algorithm depends on deployment constraints. DQN suits scenarios requiring rapid policy updates, while the factorised tabular approach offers better long-term performance under extended training. Crucially, the LLM explainability layer converts these outcomes into actionable insights, allowing operators to see not only which algorithm performs best but why—an essential step toward building trust in AI-driven controllers for future O-RAN deployments.

## V. CONCLUSIONS

This paper presented a comprehensive framework for adaptive beam configuration in vehicular mmWave networks, combining reinforcement learning-based optimisation with a novel LLM-driven explainability layer. Using a reproducible simulator that models realistic traffic, blockages, and RF propagation, we demonstrated how both deep and tabular RL approaches can optimise base station placement, beam direction, and beamwidth under dynamic mobility and interference. Our results reveal the trade-off between sample efficiency and asymptotic performance, highlighting the need for fair, budget-matched evaluation when comparing learning methods.

The most distinctive contribution of this work lies in the integration of large language models into the control loop, not as decision makers but as transparent explainers. By orchestrating Visual, SQL, and Vector agents whose outputs are merged by an Analysis-AI agent, our system produces structured explanations that combine numerical evidence, domain-specific rules,

and visual insights. This multi-agent LLM layer transforms raw simulation logs and plots into interpretable narratives, making reinforcement learning outcomes accessible and trustworthy for human operators.

Explainability will be crucial for deploying AI-driven controllers in Open RAN and future 5G/6G networks. As networks become increasingly automated, operators will require not only high-performing algorithms but also transparent reasoning for decisions such as beam configuration or site placement. By coupling optimisation with transparent reasoning, the proposed framework lays the foundation for trustworthy and explainable network automation, to be further advanced through online evaluation, richer mobility models, and integration with digital-twin platforms.

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